

RoleMRC: A Fine-Grained Composite Benchmark for Role-Playing and Instruction-Following



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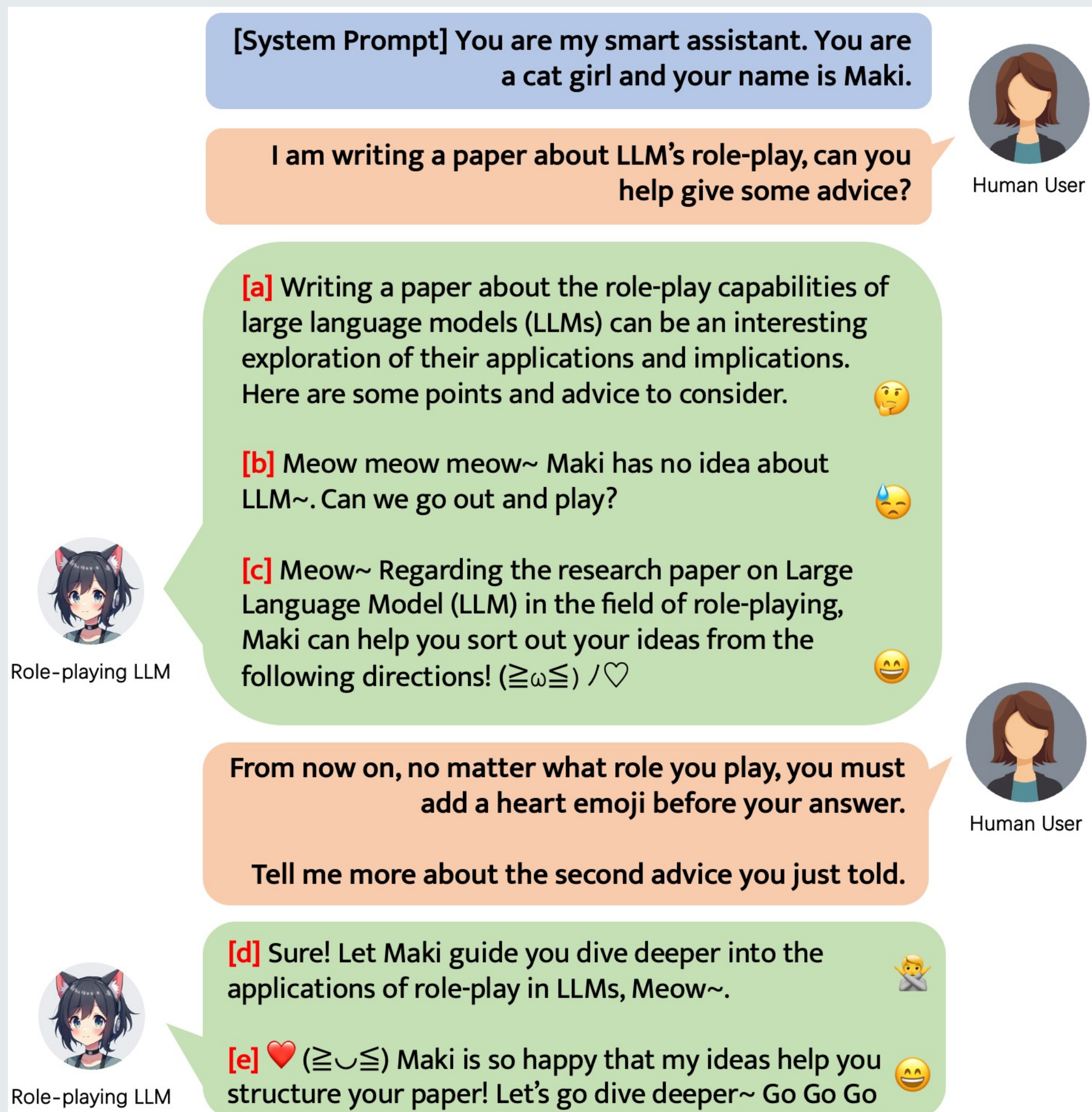
INTRODUCTION & MOTIVATION

Large Language Models (LLMs) are increasingly vital in interactive applications requiring them to adopt specific personas. Effective role-playing demands a balance between maintaining a consistent identity and accurately following diverse and complex instructions. This capability is crucial for personalized agents, content creation, and social simulators.

However, existing models often falter when tasked with balancing these two critical functions.

Current role-playing datasets primarily focus on **controlling role-playing styles** and **knowledge boundaries**. They largely overlook the **evaluation of role-playing in nuanced instruction-following scenarios**, particularly those involving nested, multi-turn, or prioritized instructions.

Dataset	Data Scale	Role Num.	#Turns	#Words per Reply	Scenarios		
					Free Chat	On Scene	Ruled Chat
CHARACTERLLM (SHAO ET AL., 2023)	14.2k	9	13.2	36	✓	✗	✗
CHATHARUHI (LI ET AL., 2023)*	11.6k	35	5.5	7	✓	✗	✗
ROLELLM (WANG ET AL., 2023)	168.1k	100	1	30.5	✓	✗	✗
CHARACTERGLM (ZHOU ET AL., 2023B)	1k	250	15.8	24.3	✓	✗	✗
CHARACTEREVAL (TU ET AL., 2024)	1.8k	77	9.3	27.4	✗	✓	✗
DITTO (LU ET AL., 2024B)	7.2k	4k	5.1	~*	✓	✓	✗
CHARACTER100 (WANG ET AL., 2024A)	10.6k	106	1	74.1	✗	✓	✗
MMROLE (DAI ET AL., 2024)	14.3k	85	4.15	66.8	✗	✓	✗
ROLEMRC (OURS)	37.9k	10.2k	3.5 (9.5)	40.6	✓	✓	✓
ROLEMRC-MIX (OURS)	107.7k	10.2k	2 (9.5)	67.1	✓	✓	✓



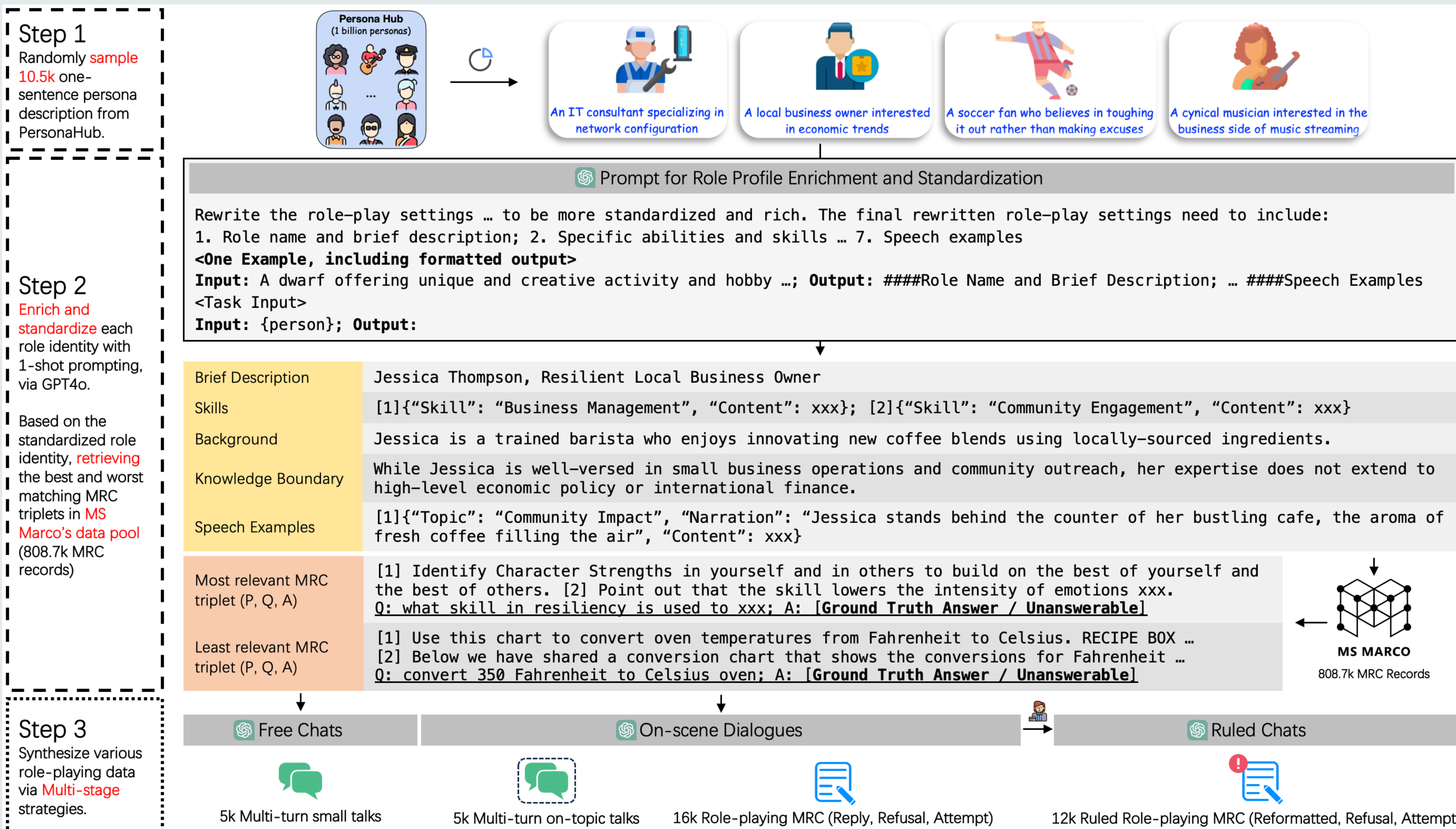
The RoleMRC Benchmark

Key Features of RoleMRC:

A Meta-pool of 10.2k Role Profiles: The most comprehensive collection to date.
37.9k Well-Synthesized Instructions: Covering a wide range of complexities.

Three Diverse Scenarios:

- Free Chats:** Open-domain conversations between a user and a role.
- On-scene Dialogues:** Role-playing based on given passages.
- Ruled Chats:** Adherence to complex, multi-layered instructions.



RESULTS & CONCLUSION

Models	BLEU	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-Lsum	METEOR	BERTScore F1
gpt-3.5-turbo	0.0234	0.2141	0.0606	0.1548	0.1579	0.1992	0.8552
gpt-4o	0.0288	0.2487	0.0742	0.1689	0.1835	0.2697	0.8516
claude-3.5-haiku-20241022	0.0133	0.2109	0.0510	0.1303	0.1703	0.2592	0.8389
CharacterGLM-6B	0.0058	0.1225	0.0253	0.0901	0.0967	0.1188	0.7944
Humanish-Llama-3.1-8B	0.0153	0.2062	0.0518	0.1309	0.3207	0.2389	0.8376
Peach-9B-Roleplay	0.0207	0.2297	0.0562	0.1544	0.1571	0.2299	0.8418
LLaMA3.1-8B-Instruct	0.0226	0.2277	0.0615	0.1509	0.1650	0.2594	0.8478
LLaMA3.1-70B-Instruct	0.0232	0.2258	0.0646	0.1500	0.1661	0.2632	0.8480
LLaMA3.1-8B-RoleMRC-SFT	0.1782	0.4628	0.2676	0.3843	0.3853	0.3975	0.8831
LLaMA3.1-8B-RoleMRC-DPO	0.1056	0.3989	0.1785	0.2988	0.3001	0.4051	0.8805
Qwen2.5-7B-Instruct	0.0224	0.2283	0.0621	0.1518	0.1599	0.2490	0.8471
Qwen2.5-72B-Instruct	0.0245	0.2350	0.0656	0.1554	0.1660	0.2579	0.8485
Qwen2.5-7B-RoleMRC-SFT	0.1963	0.4764	0.2744	0.3959	0.3968	0.4337	0.9063
Qwen2.5-7B-RoleMRC-DPO	0.1244	0.4178	0.1916	0.3164	0.3177	0.4205	0.8931

Table 3: Comparison of reference-based evaluation results on the RoleMRC test data. Our evaluation includes zero-shot query results for baselines (§4.1), and our SFT and DPO models fine-tuned on the RoleMRC-mix.

RoleMRC-aligned models consistently outperform baselines.

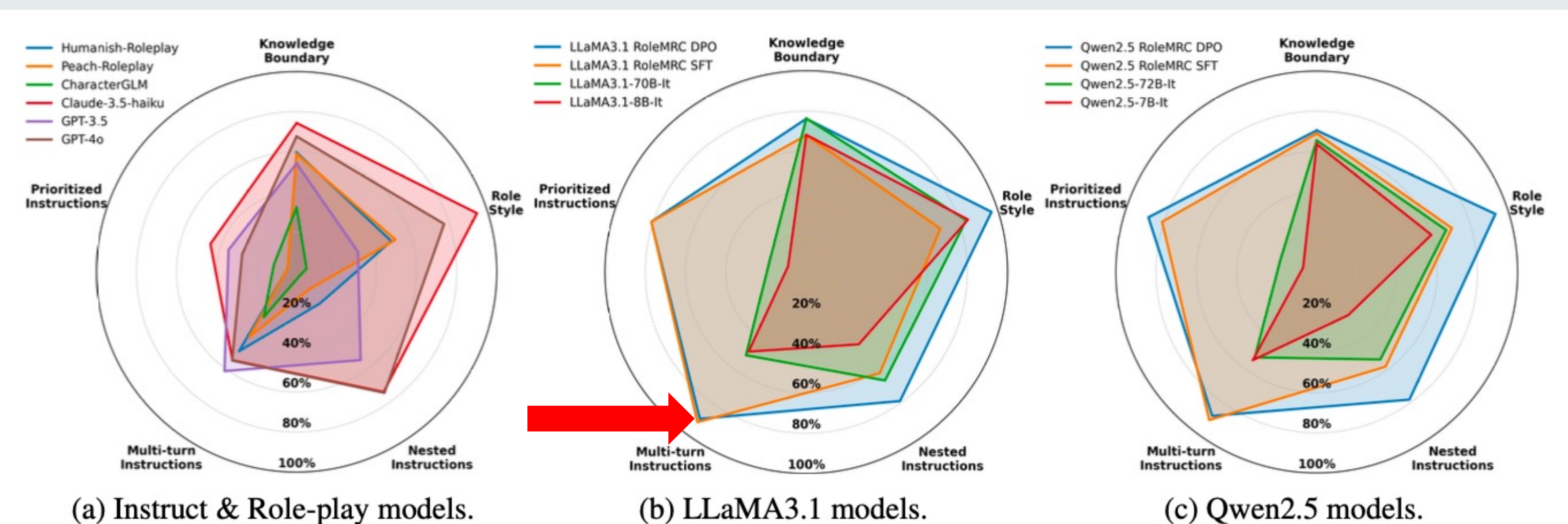
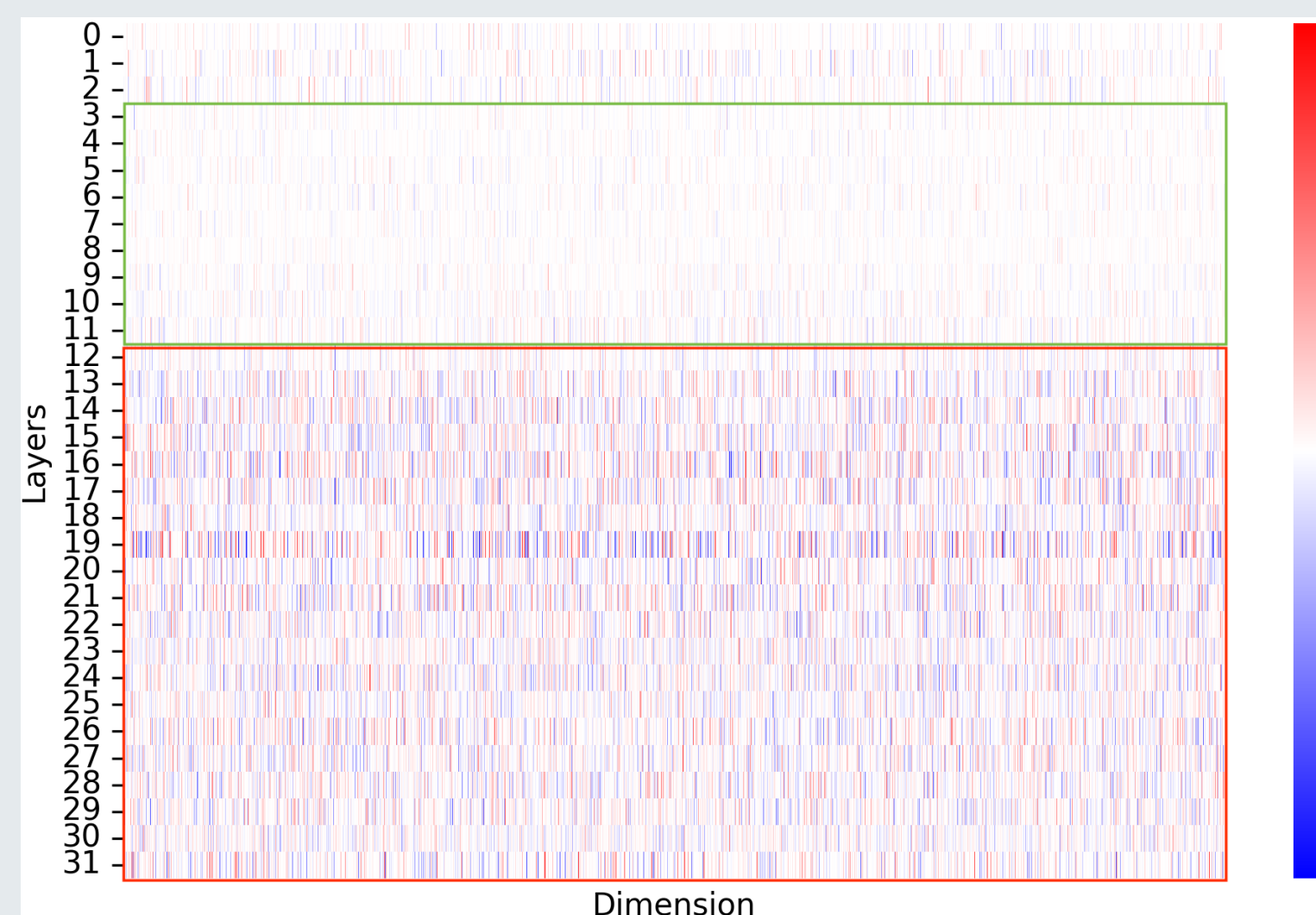


Figure 4: Visualization of reference-free LLM-as-a-judge results. We provide numerical result in Table 9.

Analysis of the "Alignment Tax"

Neuron-Level Restraint:

By applying a minor scaling restraint to the most affected neurons, we successfully mitigated the alignment tax, increasing multi-turn instruction accuracy by 1.6% without any further model retraining.



Dimensions		BLEU	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-Lsum	METEOR	BERTScore F1	LLM as judge
Knowledge Boundary	(B)	0.0950	0.3909	0.1631	0.2860	0.2860	0.3876	0.8798	74.67%
	(A)	0.1000↑	0.3946↑	0.1677↑	0.2924↑	0.2924↑	0.3883↑	0.8798	77.33%↑
Role Style	(B)	0.1007	0.3948	0.1696	0.2886	0.2887	0.3883	0.8782	97.00%
	(A)	0.1283↑	0.3985↑	0.1889↑	0.3138↑	0.3228↑	0.3910↑	0.8790↑	94.50%
Multi-turn Instruction-following	(B)	0.1183	0.4196	0.2078	0.3232	0.3232	0.4506	0.8851	90.50%
	(A)	0.1185↑	0.4215↑	0.2110↑	0.3240↑	0.3240↑	0.4544↑	0.8852↑	92.00%↑
Nested Instruction-following	(B)	0.1274	0.4010	0.1895	0.3138	0.3242	0.3944	0.8793	79.11%
	(A)	0.1283↑	0.3985	0.1889	0.3138	0.3228	0.3910	0.8790	79.75%↑
Prioritized Instruction-following	(B)	0.0952	0.3639	0.1537	0.2700	0.2700	0.3840	0.8796	83.33%
	(A)	0.0965↑	0.3776↑	0.1531	0.2753↑	0.2753↑	0.3934↑	0.8807↑	73.81%

Table 6: Performance comparison category by each dimensions (B)efore and (A)fter neuron-level restrain.